ICT2217 Network Security

Appendix A to Lab Exercise 3.1: Quick Start Guide to Supermicro Server

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Connect/Login to Supermicro server to power on Proxmox hypervisor
- Login to Proxmox hypervisor to start a pre-configured VM
- Create/Edit Linux bridge to link VM to physical port of Supermicro server
- Login to VM via Proxmox to configure IP addresses
- Graceful shutdown VM and Proxmox hypervisor

Required Hardware

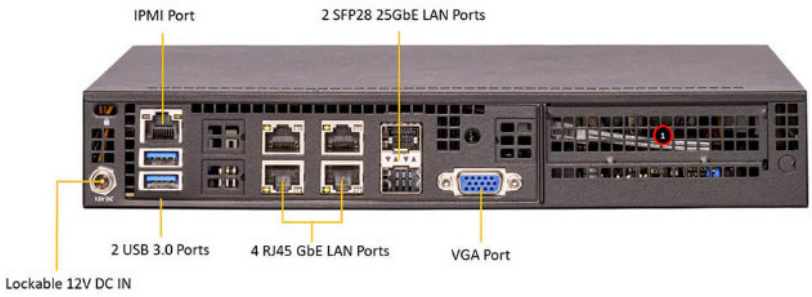
- 1 x Supermicro Server
- 1 x Ethernet cable
- 1 x Laptop

PART 0: General

0.1 For more details, please refer to Appendix B and C.

PART 1: Connecting to Supermicro Server

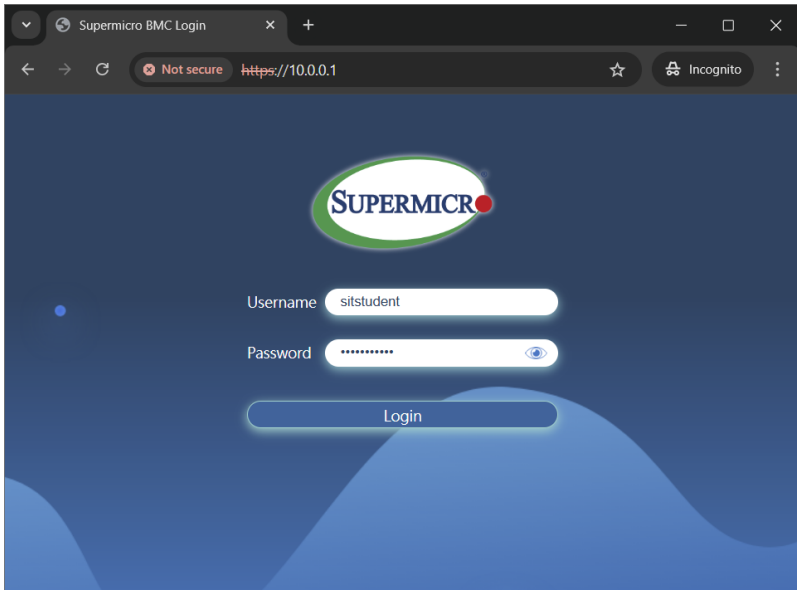
1.1 Connect a laptop using an Ethernet cable to the Supermicro server Intelligent Platform Management Interface (IPMI) port as indicated in the diagram:



	Slot Name	Slot Description
1	Slot 1	PCI-E 4.0 x16 LP

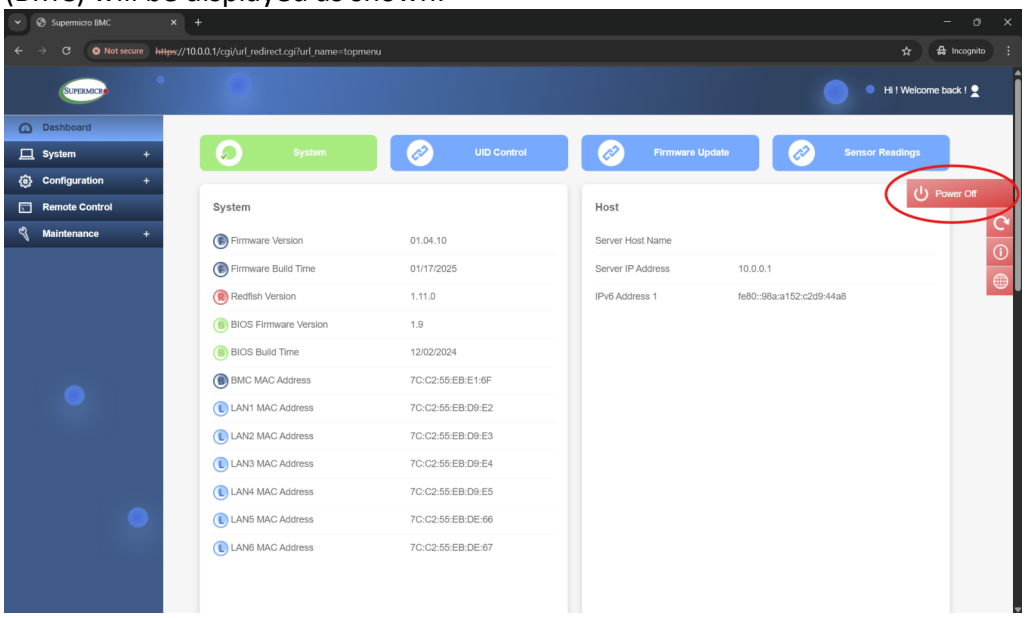
1.2 The IPMI port has been pre-configured with IP address **10.0.0.1/24**. Excluding another IP address **10.0.0.10** pre-configured for separate use later, configure any other suitable IP address on the laptop and verify that it can ping the Supermicro server.

1.3 From the laptop, start a Web browser to login to the Supermicro embedded Web server, using Username **sitstudent** as shown in the diagram and Password as written on the whiteboard in the lab.



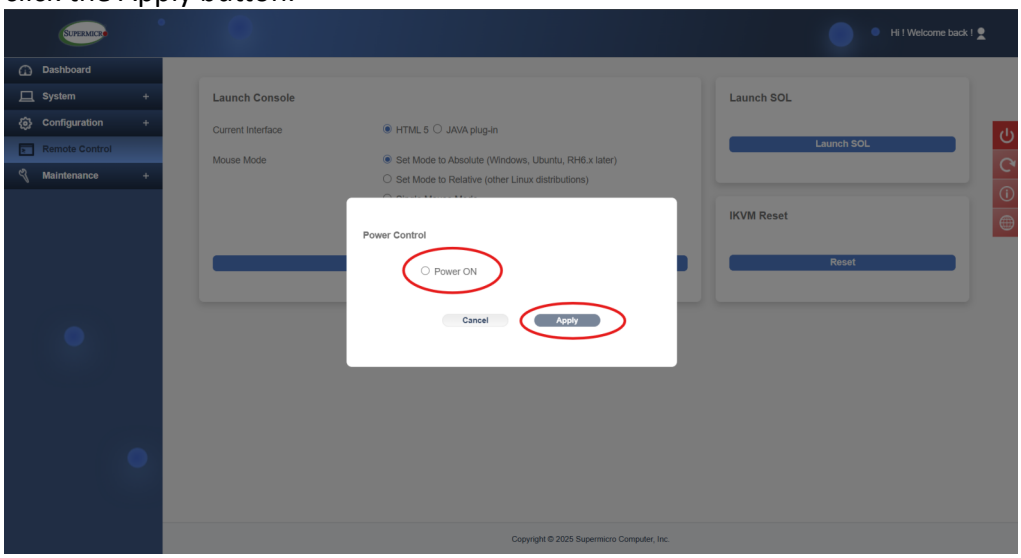
PART 2: Powering On Proxmox Hypervisor

2.1 Upon successful login, the Supermicro server Baseboard Management Controller (BMC) will be displayed as shown:



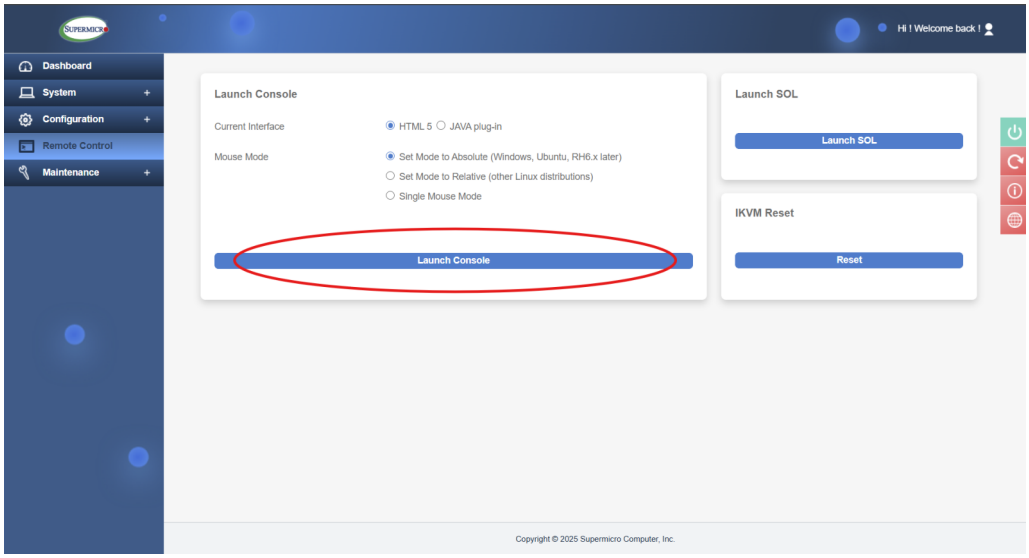
2.2 On the right of the Supermicro BMC, click the Power button to power up the Proxmox hypervisor.

2.3 A Power Control pop-up dialog box will appear. Check the Power ON option and click the Apply button.

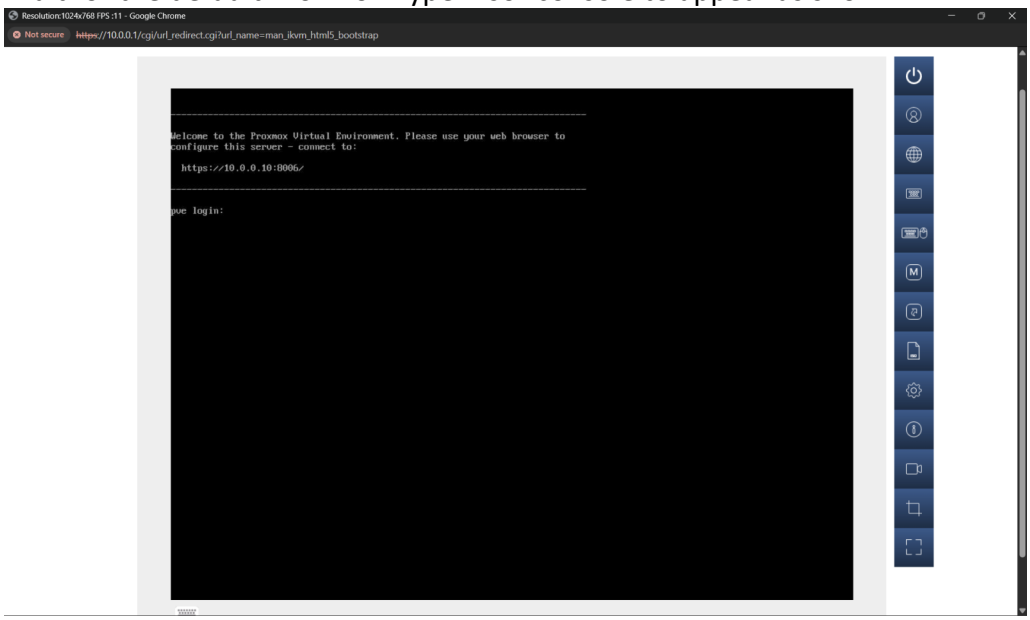


2.4 You should start hearing the fan spinning sound coming out from the Supermicro server.

2.5 After powering up, select the Remote Control tab on the left of the Supermicro BMC and click the Launch Console button:



2.6 Wait for the default Proxmox hypervisor console to appear as shown:



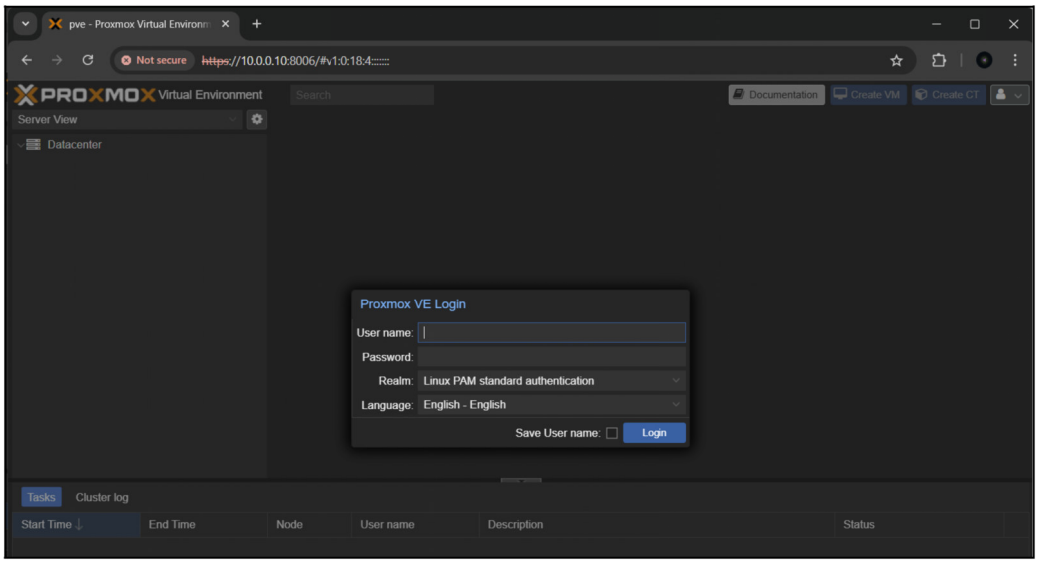
2.7 Instead of using console to manage the Proxmox hypervisor, another simpler way is to use Web UI which will be discussed next.

PART 3: Accessing Proxmox Virtual Environment

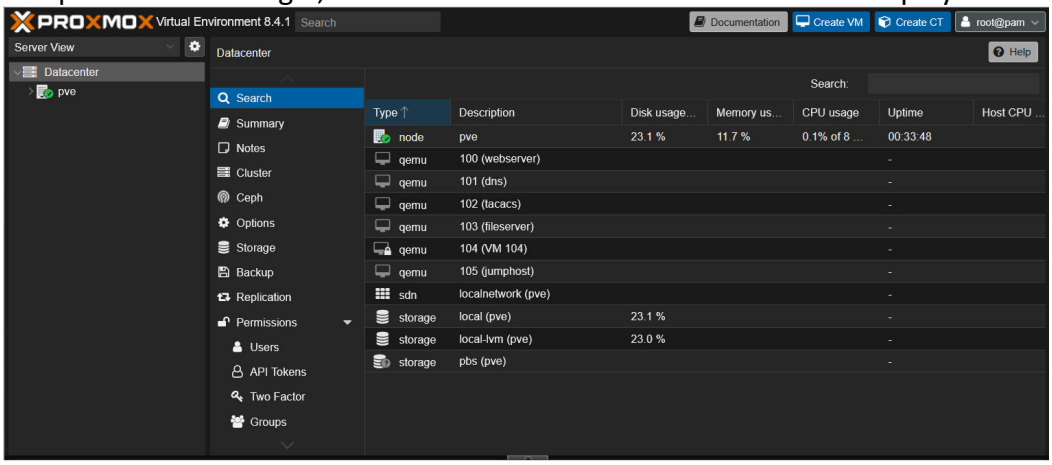
3.1 Disconnect the Ethernet cable from the Supermicro server IPMI port and connect it to LAN 1 port as shown:



3.2 The Proxmox hypervisor web UI can be accessed via URL **https://10.0.0.10:8006**, login using User name **root** and the same password as written on the whiteboard in the lab.

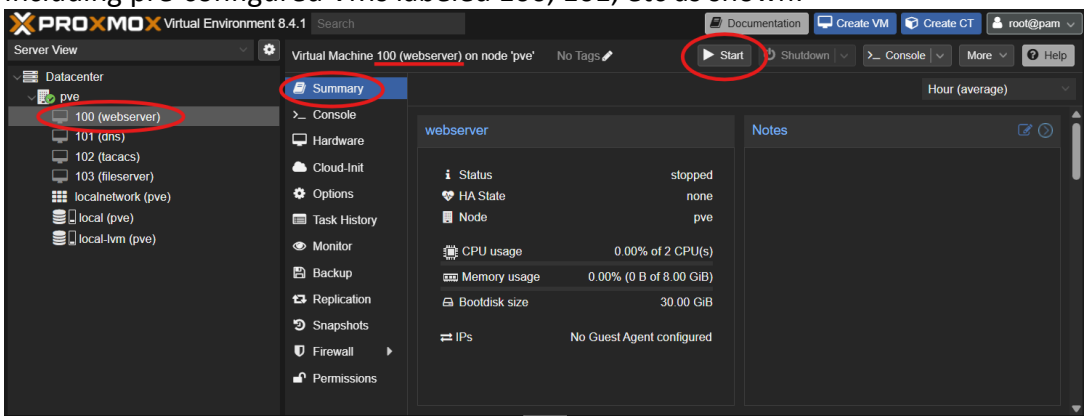


3.3 Upon successful login, the Proxmox virtual environment will be displayed as shown:



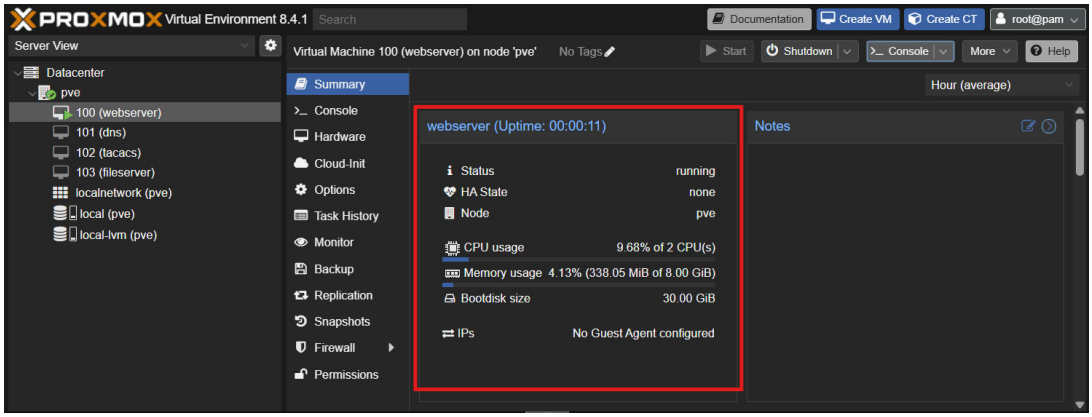
PART 4: Starting a Virtual Machine (VM) from Proxmox Virtual Environment

4.1 On the left and corresponding to Proxmox virtual environment (pve) label, click the arrow icon > which will change to downward arrow icon v with listing of details including pre-configured VMs labeled 100, 101, etc as shown:



4.2 To start a VM, select it, e.g. 100 (webserver), and click the Start button on the top-right as shown in the above diagram.

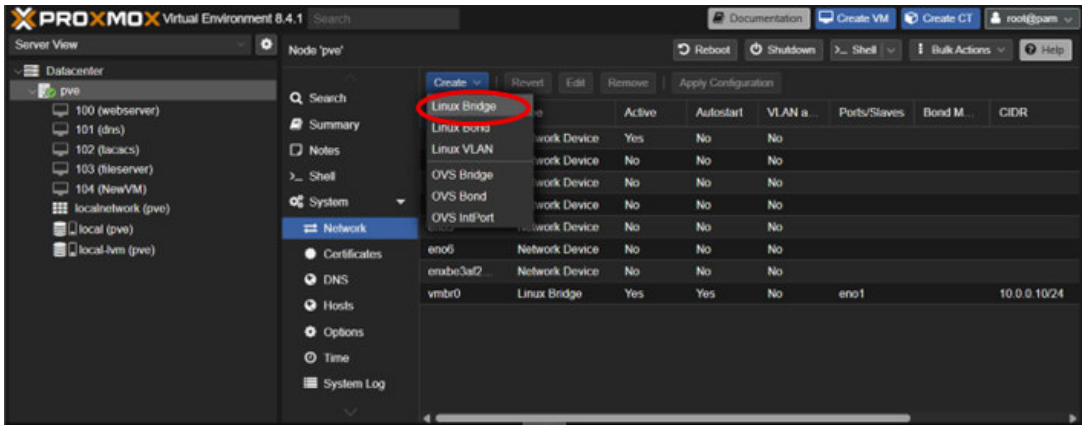
4.3 Under the Summary tab in the middle, verify that the Status is changed to running as shown:



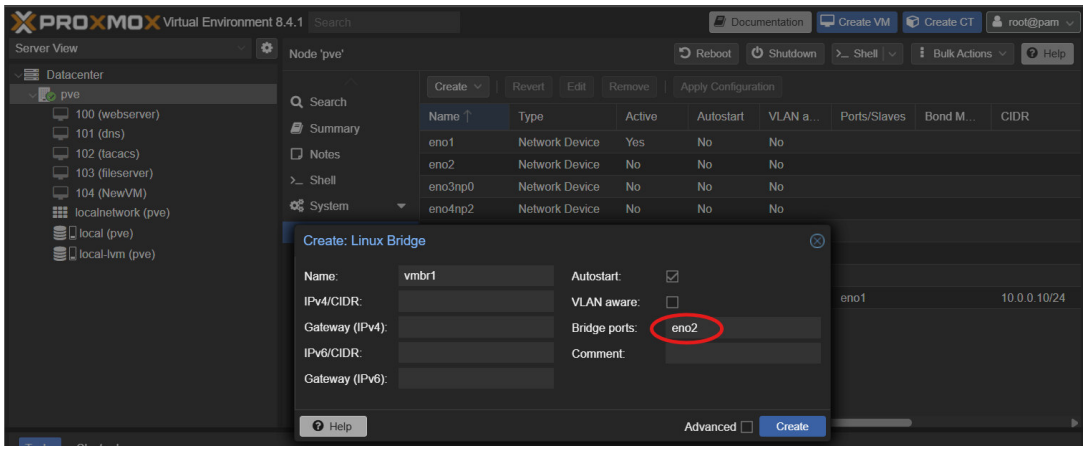
PART 5: Creating Linux Bridge and Linking to Physical Port

5.1 An intermediary Linux bridge is required to link a physical port of the Supermicro server to a VM so that the VM can be accessible in a network.

5.2 To create a Linux bridge, select pve → System → Network → Create → Linux Bridge as shown:



5.3 A Create: Linux Bridge pop-up dialog box will appear. Enter a name for the bridge, e.g. vmbr1, the Bridge port (physical port) of the Supermicro server, e.g. eno2, and click the Create button.

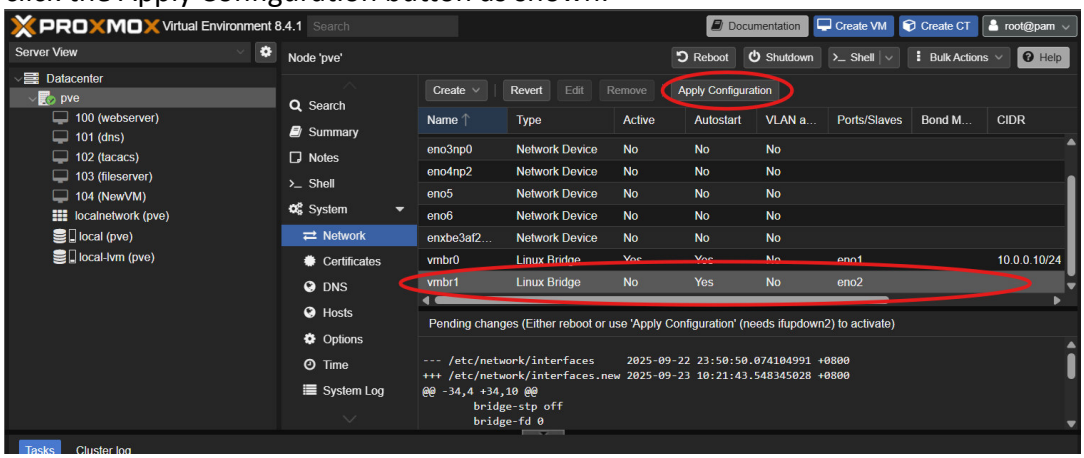


5.4 Please do NOT enter IP addresses in the dialog box which are meant for the Supermicro server, not VM.

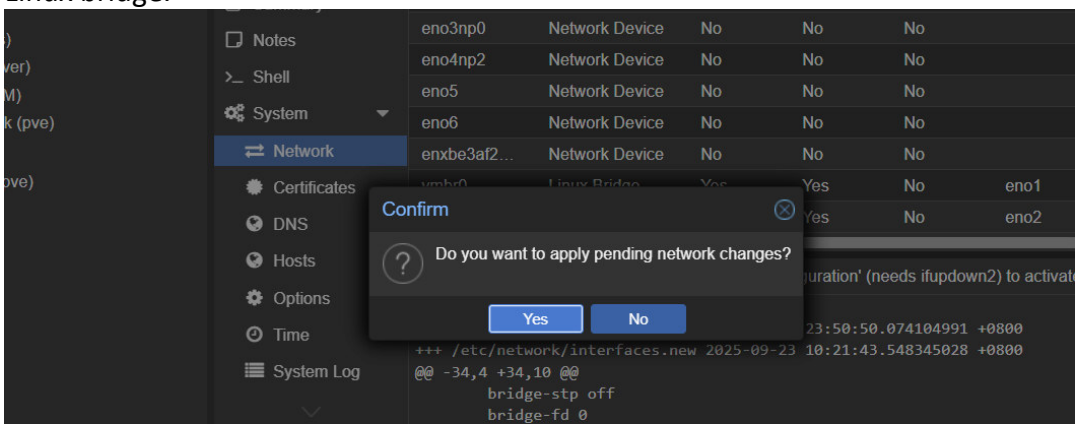
5.5 Note the mapping of the Bridge port name to physical port of the Supermicro server:

No	Name	Type	Descriptions
1	eno1	Network Device	Physical Port – LAN 1
2	eno2	Network Device	Physical Port – LAN 2
3	eno3np0	Network Device	Physical Port – SFP port 1
4	eno4np2	Network Device	Physical Port – SFP port 2
5	eno5	Network Device	Physical Port – LAN 3
6	eno6	Network Device	Physical Port – LAN 4
7	enxbe3af2b6059f	Network Device	Physical Port – IPMI
8	vmbr0	Linux Bridge	Mapped to port eno1, assigned IP 10.0.0.10/24 for Proxmox hypervisor access

5.6 Notice that the new Linux bridge is inserted but is pending changes. To activate, click the Apply Configuration button as shown:

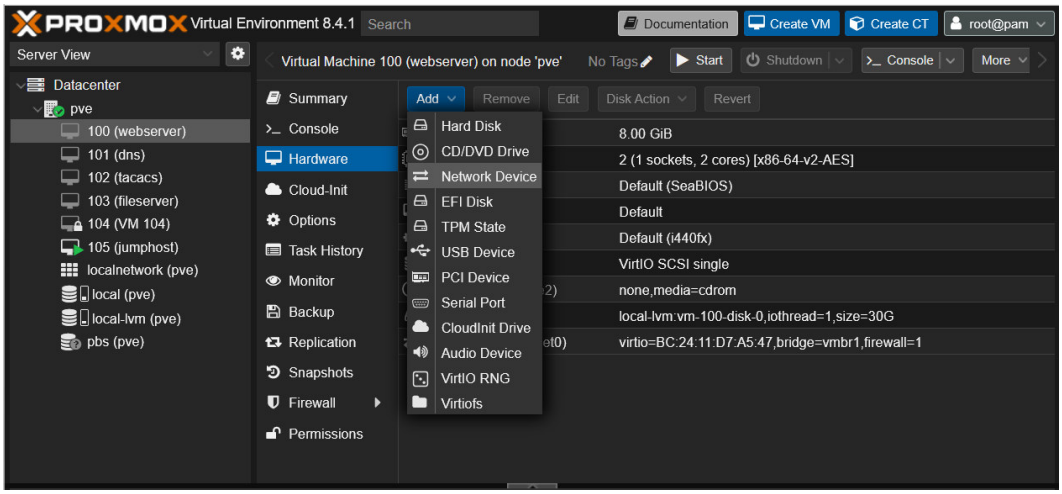


5.7 A Confirm pop-up dialog box will appear. Click Yes button to confirm creating the Linux bridge.

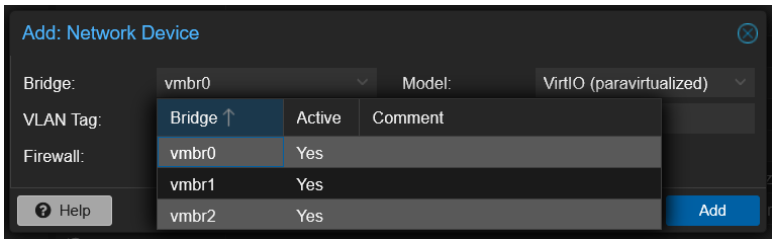


PART 6: Adding/Editing Linux Bridge to VM

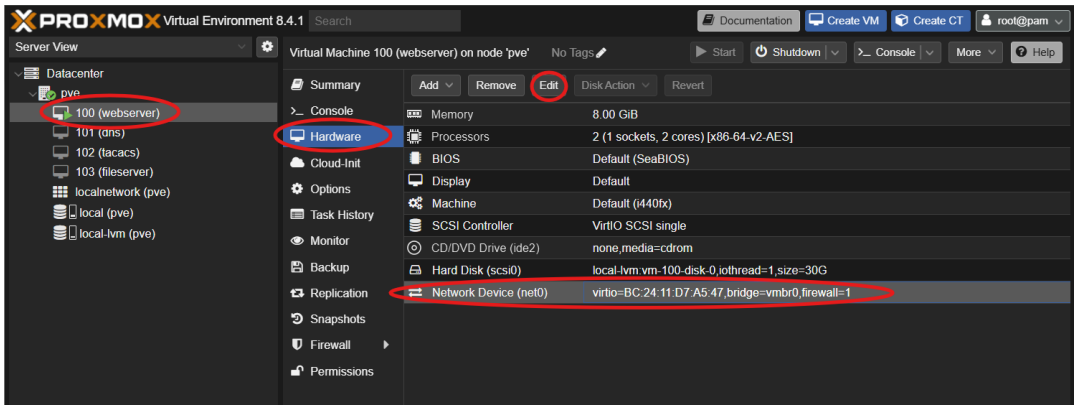
6.1 To add a Linux bridge to a VM, select the VM → Hardware → Add → Network Device as shown:



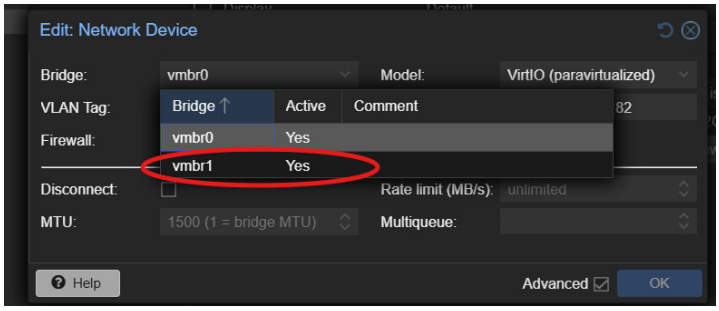
6.2 An Add: Network Device pop-up dialog box will appear for you to select which Linux bridge to add to the VM.



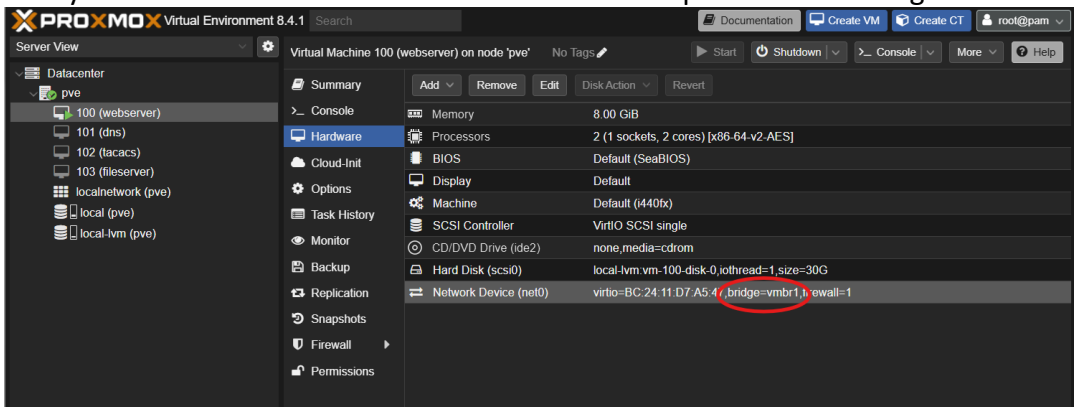
6.3 To edit a Linux bridge already linked to a VM, select the VM → Hardware → Network Device and then click the Edit button as shown:



6.4 An Edit: Network Device pop-up dialog box will appear. Click the Bridge drop-down menu and select which Linux bridge to be linked together, e.g. vmbr1, and click OK button.

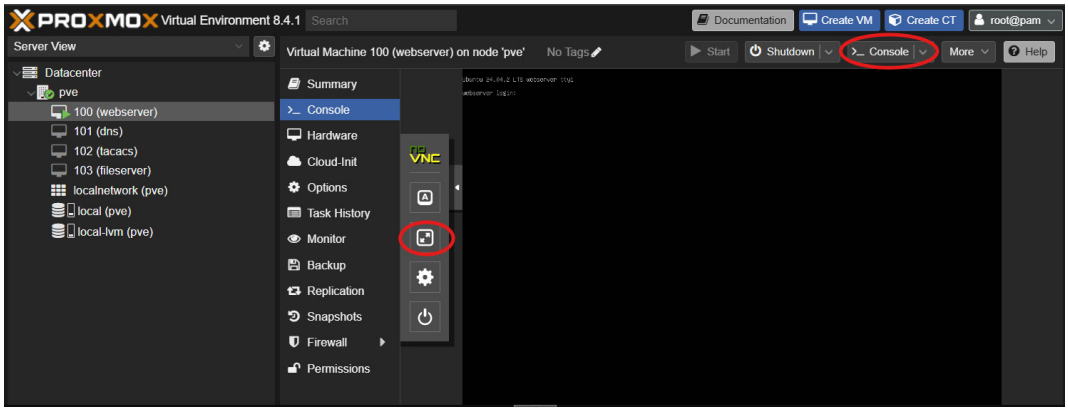


6.5 Verify that the VM Network Device is linked to the updated Linux bridge.



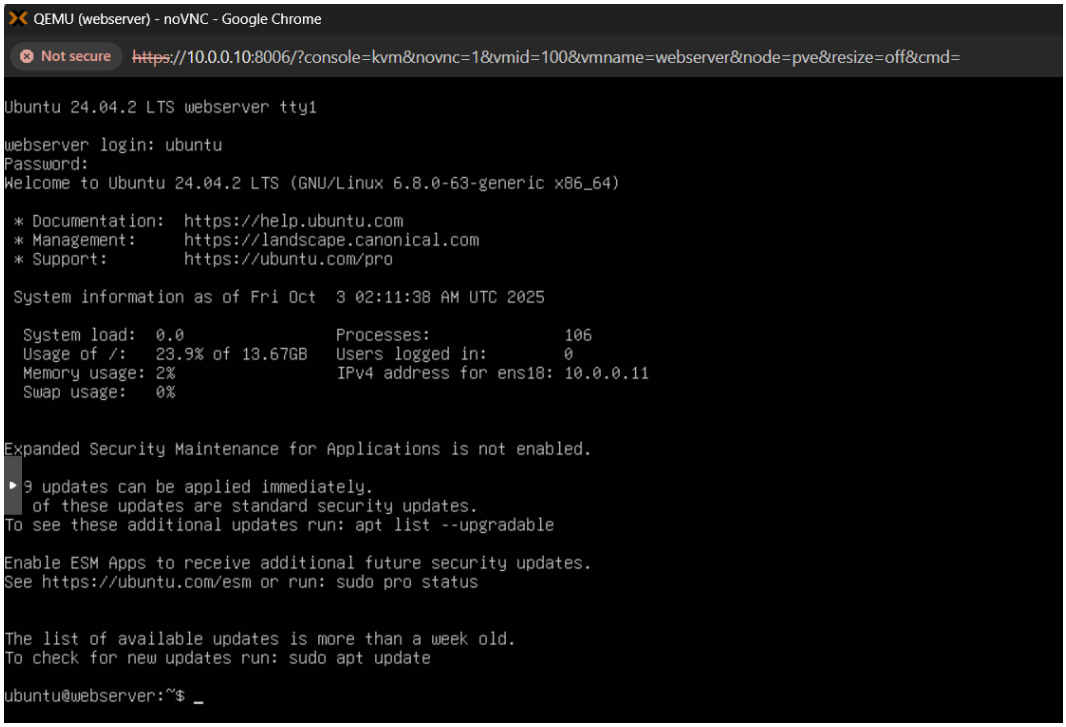
PART 7: Configuring IP addresses for a Virtual Machine (VM)

7.1 Select the VM → Console:



7.2 To have a bigger console screen, click on the full-screen icon in the middle or the Console menu on top as shown in the above diagram.

7.3 The VM is running Ubuntu Linux OS. Login using **ubuntu** and the same password as written on the whiteboard in the lab.



7.4 Change directory to /etc/netplan, and look for the default configuration filename **50-cloud-init.yaml**.

7.5 If not found, generate it using the command:

ubuntu:~$ **sudo netplan generate**

7.6 Open the configuration file **50-cloud-init.yaml** using any text editor, e.g. nano.

ubuntu:~$ **sudo nano** 50-cloud-init.yaml

```
network:
  version: 2
  ethernets:
    ens18:
      addresses:
      - "10.0.0.15/24"
      nameservers:
        addresses:
        - 8.8.8.8
        search: []
      routes:
      - to: "default"
        via: "10.0.0.254"
    ens19:
      addresses:
      - "20.0.0.1/24"
```

7.7 Edit the IP addresses of VM interface ens18 linking to vmbr1 mapping to eno2, and VM interface ens19 linking to vmbr2 mapping to eno5 if applicable. Remember to save your changes.

7.8 Apply the changes using the command:

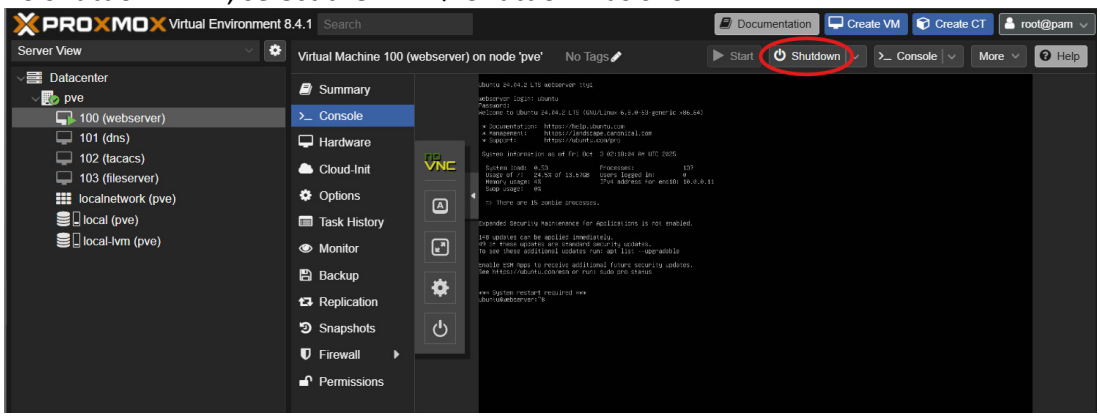
ubuntu:~$ **sudo netplan apply**

7.9 Connect up your network to the correct physical port(s) of the Supermicro server. Verify that the VM is now accessible via the configured IP addresses.

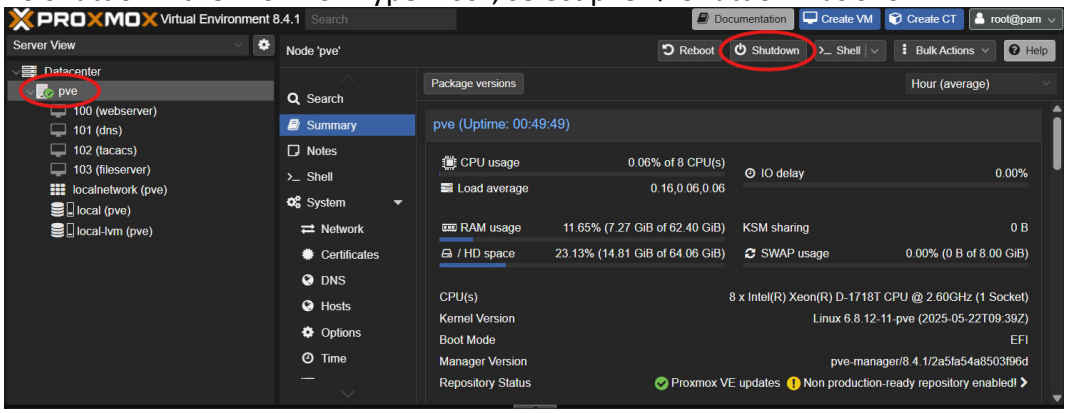
PART 8: Shutting Down VM and Proxmox Hypervisor

8.1 **Important!! Please help to take care of the Supermicro server by performing graceful shutdown of the VMs and Proxmox hypervisor before powering down the network rack.**

8.2 To shutdown VM, select the VM → Shutdown as shown:

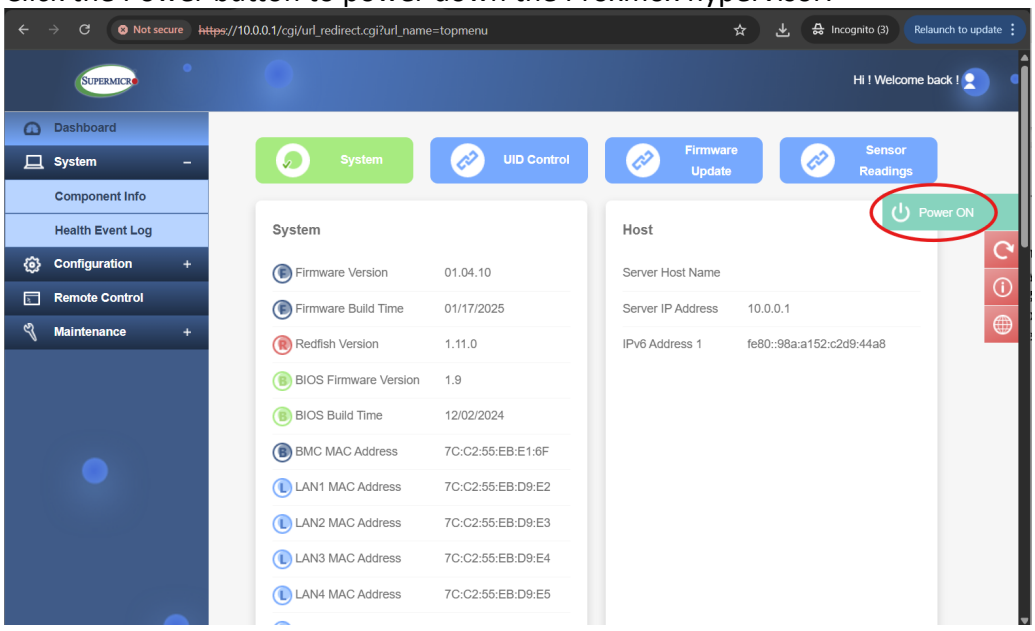


8.3 To shutdown the Proxmox hypervisor, select pve → Shutdown as shown:

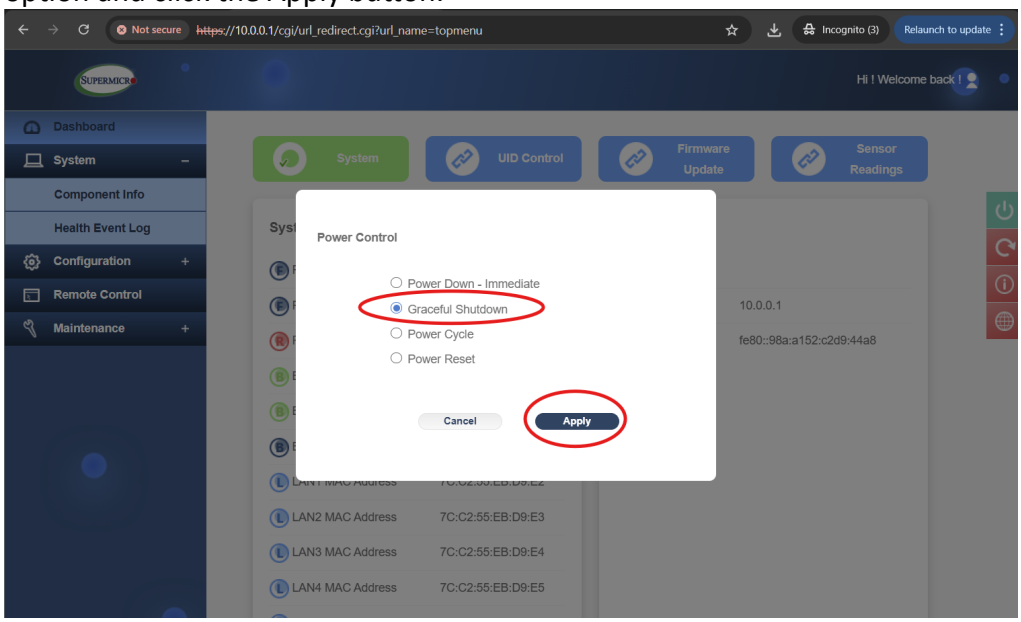


8.4 Alternatively, disconnect the Ethernet cable from LAN 1 port and reconnect it back to IPMI port.

8.5 Click the Power button to power down the Proxmox hypervisor:



8.6 A Power Control pop-up dialog box will appear. Check the Graceful Shutdown option and click the Apply button.



8.7 You should stop hearing the server fan spinning sound, and also able to feel that not much air is flowing out through the fan outlet slot grill as shown:

